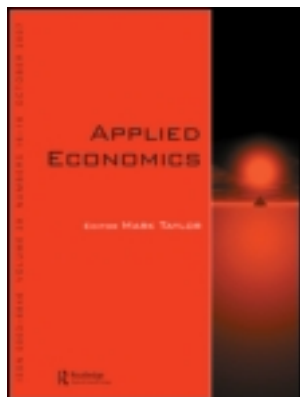




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Efficiency and risk in Taiwan banking: SBM super-DEA estimation

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In this study we adopt the two Data Envelopment Analysis (DEA) methods, including Banker–Charnes–Cooper (BCC) and Slacks-based Measure (SBM) super efficiency, to investigate whether a bank's technical efficiency index's results, incorporating account risk and not incorporating account risk, differ significantly. The information is obtained from 43 Taiwanese banks for the period 1998 to 2002. We also employ the nonradial and slacks-based Malmquist Total Factor Productivity (TFP) index to measure the impact of productivity change on the panel data. Our empirical results from the DEA approach are summarized as follows: (1) Loan quality factors have an impact on a bank's efficiency. In general, the bank with a higher degree of problem loans drops down its efficiency by incorporating account risk; however, the bank with a lower degree of problem loans increases its efficiency by incorporating account risk. (2) Super efficiency provides a good framework for ranking efficient units. (3) Based on the nonradial and slacks-based Malmquist TFP index, we find that bank productivity has increased and technical change has improved. (4) In Taiwan, bank with a higher performance will be more secure.

1. Introduction

In the early 1990s, Taiwan began to follow the general trend towards bank deregulation in order to increase operating efficiency and to attract funds into the loanable fund supply market. Deregulation in the financial market is now perceived as a worldwide phenomenon and Taiwan is no exception to this international trend. The policy of deregulation involves two distinct streams: privatization of ownership and the establishment of new banks in the market. In 1991 the government in Taiwan announced the Commercial Bank Establishment Promotion Decree as a means of opening up its

bank market further by inviting private investors to participate in Taiwan's banking industry. The investors set up 25 new commercial banks, pushing the total number of domestic commercial banks in Taiwan in 1999 to 41.

The Bank of Taiwan provides the authority to open a new bank or a new branch, if certain prerequisites on capital adequacy are met. The implementation of such a policy clearly signals Taiwan's desire to make banking more competitive and efficient for all banks. However, although increased competition can encourage banks to operate more efficiently, it also prompts banks to take on more risk. Due to the rapid development of

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financial markets and financial instruments, banks are now possessing more options while their assets are also exposed to excessively high risks, especially in bank loans. Until the end of 2000, the average ratio of problem loans for all banks went up to 8.16%. In 2000 the government announced a bank law amendment for bank financial supervision. A year later, the government announced the capital adequacy requirement law with a target to drop the ratio of problem loans to 5% and to increase capital adequacy to 8%.

From the above competitive pressure and the government's current regulation policy within Taiwan's banking industry, we know that a higher level of competition not only can improve a bank's efficiency, but also pushes banks towards higher risk. Hence, it is important to evaluate bank efficiency by incorporating account risk at a higher level of competition. Thus, not only will banks be efficient, but they also can have less risk. For this reason, efficiency and risk are two aspects that must occur jointly together.

Most studies to date on bank efficiency have focused mainly upon the economies of scale and scope (e.g. Hunter and Timme, 1986; Berger *et al.*, 1987; Ferrier and Lovell, 1990; Berger and Humphrey, 1991; McAllister and McManus, 1993), total productivity (e.g. Aly *et al.*, 1990; Grabowski *et al.*, 1993; Favero and Papi, 1995; Zaim, 1995; Schaffnit *et al.*, 1997; Fukuyama *et al.*, 1999), comparing parametric and nonparametric programming approaches (e.g. Ferrier and Lovell, 1990; Bhattacharyya *et al.*, 1997; Resti, 1997; Chen and Yeh, 2000; Huang and Wang, 2002), and the efficiency effect (e.g. Cebenoyan *et al.*, 1993; Barr *et al.*, 1994; Elyasiani *et al.*, 1994; Favero and Papi, 1995; DeYoung and Hasan, 1998; Chang, 1999; Casu and Molyneux, 2003).

Due to the rapid development of financial markets and competition resulting in a high nonperforming ratio for troubled banks, many research studies have adopted risk into banks' efficiency measure. Berger and DeYoung (1997) summarize four hypotheses, including 'bad luck' hypotheses, 'bad management' hypotheses, 'skimping behaviour' hypotheses and 'moral hazard problems' hypotheses. Except for moral hazard problems, the hypotheses do not describe the relationship between measured efficiency and problem loans. Three hypotheses have discussed the relationship between measured efficiency and problem loans. Under 'bad luck' hypotheses, increasing nonperforming loans in banks will result in decreasing bank efficiency since banks will purchase the additional input to solve those problem loans. In the 'bad management' hypotheses, a low measured efficiency is a signal of poor management. As for bad

management, they may have difficulty monitoring and controlling the borrowers after loans, which can be expected to increase problem loans. In the 'skimping' hypotheses, the amount of resources allocated to monitoring loans affects loan quality and efficiency. All four hypotheses show that problem loans are an important factor to measure banking efficiency.

There are two issues concerning bank efficiency and risk. One treats risk them as exogenous in order to analyse efficiency effects (e.g. Cebenoyan *et al.*, 1993; Barr *et al.*, 1994; Elyasiani *et al.*, 1994; Berger and DeYoung, 1997; Ataullah *et al.*, 2004). The above results show that the efficiency level is significantly correlated with the risk indicators. The other issue treats risk as endogenous in order to analyse bank efficiency (e.g. Mester, 1996; Chang, 1999; Hughes, 1999; Pastor, 1999; Altunbus *et al.*, 2000; Hughes *et al.*, 2000; Hughes *et al.* 2001; Drake and Hall, 2003; Girardone *et al.*, 2004). They show that if risk and factors are taken into account, then the risk factors have the biggest influence on the efficiency estimates and ranking. They also present that increasing risk (problem loans) will tend to decrease efficiency.

One problem does happen when the nonperforming ratio is considered as the risk index of banks. Since Taiwanese domestic banks are small in scale and there are many of them, by using the traditional Data Envelopment Analysis (DEA) (Banker, Charnes and Cooper (BCC) or Charnes, Cooper and Rhodes (CCR)) to estimate the efficiency score for these banks, the score will turn out many efficient Decision Making Units (DMUs) and create a ranking problem. In other words, it will be unable to rank and compare the efficiency of the performances correctly. Although Andersen and Petersen (1993) propose a super-efficiency model to solve the ranking problem, the super-efficiency model has an infeasible problem. In order to deal with the ranking of Taiwan's banking problem and infeasible problems, we use the Tone (2002) Slacks-based Measure (SBM) super-efficiency model and the nonradial and slacks-based Malmquist Total Factor Productivity (TFP) model to deal with ranking the efficient decision making units and infeasible problems.

The purpose of our analysis is two-fold. First, this study uses the BCC and slack-based super-efficiency model approaches to investigate whether a bank's technical efficiency index results, incorporating account risk and not incorporating account risk, differ significantly. Secondly, we also employ the nonradial and slacks-based Malmquist TFP index to measure the impact of productivity change on the panel data.

This article is organized as follows: the first section provides the motivation for the research; the second section is a literature review; the third section offers the introduction to the methodologies; the fourth section shows the empirical results; the final section presents the conclusions.

II. Literature Review

To date, research on bank efficiency has focused mainly upon the economies of scale and scope, total productivity and the efficiency of banks by comparing the parametric and nonparametric programming approaches and the efficiency effect. In regards to the economies of scale and scope, Hunter and Timme (1986), Berger *et al.* (1987), Ferrier and Lovell (1990), Berger and Humphrey (1991) and McAllister and McManus (1993) focus on estimating the functional characteristics and economies of scale and scope. The research on the total productivity and efficiency of banks includes Aly *et al.* (1990), Grabowski *et al.* (1993), Favero and Papi (1995), Zaim (1995), Schaffnit *et al.* (1997) and Fukuyama *et al.* (1999). Ferrier and Lovell (1990), Bhattacharyya *et al.* (1997), Resti (1997), Chen and Yeh (2000) and Huang and Wang (2002) examine bank technical efficiency by comparing the parametric and nonparametric programming approaches.

Cebenoyan *et al.* (1993), Barr *et al.* (1994), Elyasiani *et al.* (1994), Favero and Papi (1995), DeYoung and Hasan (1998), Chang (1999) and Casu and Molyneux (2003) change their focus from a one-stage approach to the two-stage approach in analysing efficiency effects. Favero and Papi (1995) and DeYoung and Hasan (1998) measure allocative efficiency and technical efficiency by analysing a bank-specific measure of inefficiency in order to investigate the determinants of a bank's efficiency.

Researchers have recently focused on the relationship between bank efficiency and risk when studying a bank's efficiency. There are two issues concerning bank efficiency and risk. One treats risk as exogenous in order to analyse efficiency effects, while the other treats risk as endogenous in order to analyse bank efficiency.

For efficiency effects, Cebenoyan *et al.* (1993), Barr *et al.* (1994), Elyasiani *et al.* (1994), Berger and DeYoung (1997) and Ataullah *et al.* (2004) apply the two-stage approach to analyse efficiency effects in terms of incorporating risk effects, including nonperforming loans, allowance for loan losses, and risky assets. The first stage calculates a set of efficiency indices. The second stage uses risk indicators,

including nonperforming loans, allowance for loan losses and risky assets to explain the efficiency score. The above results show that the efficiency level is significantly correlated with the risk indicators.

Mester (1996) points out that 'unless quality and risk are controlled for, one might easily miscalculate a bank's level of inefficiency'. Mester uses Third Federal Reserve District data to seek the efficiency of banks, accounting for the quality and riskiness of bank output. Mester identifies that banks in the US operate at cost efficient output levels.

Chang (1999) incorporates risk to measure the technical efficiency of financial intermediaries in Taiwan. Chang finds that incorporating risk as an undesirable output has a significant influence on the technical efficiency estimates. Pastor (1999) proposes a new idea to break down the banking risk-provision for loan losses into internal and external components in order to obtain a bank's efficiency adjusted risk. The result shows that the deregulation of Spain's banking system has affected banks' conduct in terms of efficiency and risk.

Hughes (1999) shows the importance of incorporating endogenous risk into the analysis of production and measures banks' profit efficiency and scale economies in US commercial banking by incorporating risk and financial structure in the model. Hughes *et al.* (2000) bridge the gap by incorporating risk into a bank production model so that this model can analyse risk-based phenomena such as diversification and moral hazard. They use data on US commercial banks to illustrate that there are significant differences between their generalized model and the standard profit-maximization model, which ignores risk.

Altunbas *et al.* (2000) take data on Japanese banking from 1993 to 1996 to test the impact of risk and quality factors on banks' cost by using the stochastic cost frontier methodology. They find that optimal bank size is considerably smaller when risk and quality factors are taken into account. They also see that the level of financial capital has a significant influence on the scale efficiency estimates. Altunbas *et al.* (2000) focus on the impact of risk and output quality factors on banks' cost efficiency and technical change in Japanese banks, where OBS items are included as output. They show that if risk and quality factors are not taken into account, then optimal bank size tends to be overstated. They also offer that the level of financial capital has the biggest influence on the scale efficiency estimates.

Hughes *et al.* (2001) explore how to incorporate banks' capital structure and risk taking into models of production. They take US data in 1994 to seek the scale efficiency of banks. They find that the levels of

financial capital and risk-taking have significant influences on the scale efficiency estimates.

Drake and Hall (2003) use data envelopment analysis to estimate technical and scale efficiency in Japanese banking. This article, using problem loans as an exogenous variable, discusses problem loans' influence on bank efficiency. The result is that problem loans are an important factor to impact Japanese banking efficiency, especially for small regional banks.

Girardone *et al.* (2004), including quality and risk of bank outputs, employ a fourier-flexible stochastic cost frontier to estimate Italian banks' cost efficiency. The result shows that the mean X-inefficiencies range between 13 and 15% of total costs. They present that the inclusion of risk and output quality variables in the cost function seems to reduce the significance of scale economy estimates.

From the above review, (summary in Appendix) we know that most researchers do not consider ranking the efficient DMUs and infeasibility problem in the super-efficiency DEA model. Thus this article uses the SBM super-efficiency model and the nonradial and slacks-based Malmquist TFP model to evaluate the efficiency and productivity of banks.

III. Data Source and Methodologies

Data source and limitation

This study uses the panel data of 43 Taiwanese commercial banks for the 3-year period from 1998 to 2002 and examines a bank's efficiency by using DEA programming (super efficiency). The data of 43 commercial banks include annual reports released by the Securities and Futures Commission (SFC) of the Republic of China. Table 1 describes the pertinent

Table 1. Definition and explanation of variables

Variable	Definition
Y_1	Loan services (including business loans and individual loans) (NT\$ billion)
Y_2	Portfolio investments (mainly in government securities and shares, along with public and private enterprises' securities)
L	Number of bank employees
M	Total deposits
K	Net book value of fixed assets
Risk	Problem bad loans (as a percentage of total loans)

Source: Reports released by the SFC of the ROC and the Taiwan Economic News Service.

definitions of the variables, as well as the original data and explanations of the data.

This study includes two types of banking output: loan services (including business loans and individual loans) and portfolio investments (such as marketable securities issued by the government, public enterprises and/or private enterprises). There are four types of input: the number of bank employees, assets, deposits and risk. Bank deposits are defined as the sum of checking accounts, savings deposits and time deposits. There is one risk indicator: nonperforming loans to present risk which follows the approach of Charnes and Cooper (1990) in incorporating nonperforming loans as an input in the DEA analysis.

Methodologies

This article attempts to extend the DEA approach to characterize a bank's technical efficiency. The DEA approach was proposed by Charnes *et al.* (1978) and developed further by Banker *et al.* (1984). CCR efficiency concept is subject to the strong hypothesis of Constant Returns to Scale (CRS). Subsequent papers have considered alternative sets of assumptions, such as BCC, who propose a variable returns to scale model. The BCC model can determine the returns to scale (increasing, constant, or decreasing Returns to Scale (RTS)) for each of the DMUs. Seiford and Thrall (1990) offer basic DEA models in input-based and output-based versions. Tone (2001) introduces a slacks-based measure of efficiency (SBM) in order to make up for the CCR model deficiency, whereby CCR models evaluating the radial efficiency do not include the nonzero slacks.

Andersen and Petersen (1993) provide super-efficiency DEA models, providing the rank of the efficient DMUs relative to a reference technology spanned by all other units. However, Thrall (1996), Zhu (1996), Dula and Hickman (1997), Seiford and Zhu (1999), Xue and Harker (2002), Tone (2002), Lovell and Rouse (2003) and Bogetoft and Hougaard (2004) show the infeasible problem in the super-efficiency model. Dealing with infeasible problems in the super-efficiency model has indeed become an important issue.

This study uses DEA, including BCC and SBM super-efficiency estimation to investigate a bank's technical efficiency indices. We also employ the nonradial and slacks-based Malmquist TFP index to measure the impact of productivity change on the panel data. The BCC, SBM super-efficiency model, and nonradial and slacks-based Malmquist TFP are as follows.

Banker–Charnes–Cooper. The piecewise linear frontiers that are constructed by employing mathematical programming techniques were introduced by Charnes *et al.* (1978) and extended by Banker *et al.* (1984). The basic model primarily utilizes the input and output quantities to measure bank efficiency.

In the Banker *et al.* (1984) model, the sets of DMUs are as I , for each DMU $i \in I$. Let us define the following variables: Y_{in} is the n -th output of the i -th DMU, X_{im} is the m -th input of the i -th DMU, λ_i is the weight of the i -th DMU, and W_{im}^- and W_{in}^+ are the input slacks and the output slacks, respectively. Here a DMU uses M inputs to produce N outputs; θ_i is the score of the i -th DMU. Term ε is a small (non-Archimedean) positive number. The input-oriented BCC method used to calculate technical efficiency can be formulated as:

$$\begin{aligned} \min \theta_k^* \\ \text{s.t.} \quad & - \sum_{i \in I} \lambda_i X_{im} + \theta_k X_{km} - W_{km}^- = 0, \quad \forall m \in M \\ & \sum_{i \in I} \lambda_i Y_{in} - W_{kn}^+ = Y_{kn}, \quad \forall n \in N \\ & \sum_{i \in I, i \neq k}^R \lambda_i = 1 \quad \lambda_i \geq 0, \quad \forall i \in I \end{aligned} \quad (1)$$

Super efficiency. In recent years, super-efficiency DEA models have become an interesting research subject that discriminates between these efficient DMUs. Lovell and Rouse (2003) point out that many papers have been proposed for super-efficiency models. Charnes *et al.* (1992), Charnes *et al.* (1996), Zhu (1996) and Seiford and Zhu (1998) utilize them to study the sensitivity of the efficiency classifications. Thrall (1996) and Dula and Hickman (1997) classify DMUs into extreme-efficient and nonextreme efficient groups for the super-efficiency model. Andersen and Petersen (1993) propose their use in ranking efficient DMUs. Rousseau and Semple (1995) apply them in two-person ratio efficiency games. Wilson (1995) and Thrall (1996) employ them to identify outliers in the data. Fare *et al.* (1994) use them to calculate and decompose the Malmquist productivity index. However, Thrall (1996) notes that the super-efficiency CRS model may be infeasible. Zhu (1996) presents that infeasibility occurs if and only if input (or output) values exist with a zero in the CRS super-efficiency model. Dula and Hickman (1997), Seiford and Zhu (1999) and Xue and Harker (2002) show the infeasible problem in the variable RTS super-efficiency model.

They also provide necessary and sufficient conditions for infeasibility of the RTS super-efficiency models. To deal with infeasible problems, Tone (2002) offers a slacks-based model, Lovell and Rouse (2003) use an equivalent standard DEA model and Bogetoft and Hougaard (2004) have theirs based on potential slack to evaluate the super-efficiency scores.

This article herein takes the SBM super-efficiency mode to overcome the infeasible problems. The SBM super-efficiency models are as follows.

SBM super-efficiency model. Following the Tone (2002) model, let us define the following variables: Y_{in} is the n -th output of the i -th DMU, X_{im} is the m -th input of the i -th DMU and the production possibility set $P(X_0, Y_0)$ spanned by (X, Y) excluding (X_0, Y_0) .

$$P(X_0, Y_0) = \left[(\bar{X}, \bar{Y}) \mid \bar{X} \geq \sum_{i \in I, i \neq 0}^n \lambda_i X_i, \bar{Y} \leq \sum_{i \in I, i \neq 0}^n \lambda_i Y_i, \bar{Y} \geq 0, \lambda \geq 0 \right] \quad (2)$$

We define a subset $\bar{P}(X_0, Y_0)$ of $P(X_0, Y_0)$ as:

$$\bar{P}(X_0, Y_0) = (X_0, Y_0) \cap [\bar{X} \geq X_0 \text{ and } \bar{Y} \leq Y_0] \quad (3)$$

$\bar{P}(X_0, Y_0)$ is not empty and $X > 0, Y > 0$.

From Equation 1, we know that the degree of super efficiency of (X_0, Y_0) under estimation is excluded from the reference set. In other words, the super-efficiency DEA models are based on reference technology constructed from all other DMUs. Thus, the SBM super-efficiency method used to calculate technical efficiency can be formulated as:

$$\begin{aligned} \phi^* = \min \phi &= \frac{(1/s) \sum_{m=1}^s X_m^- / X_{m0}}{(1/t) \sum_{n=1}^t \bar{Y}_n / Y_{n0}} \\ \text{Subject to} \quad & \bar{X} \geq \sum_{i \in I, i \neq 0}^n \lambda_i X_i \\ & \bar{Y} \leq \sum_{i \in I, i \neq 0}^n \lambda_i Y_i \\ & \bar{X} \geq X_0 \quad \text{and} \quad \bar{Y} \leq Y_0 \\ & \bar{Y} \geq 0, \quad \lambda \geq 0 \end{aligned} \quad (4)$$

Equation 4 can be restructured as the following linear programming form:

$$\gamma^* = \min \gamma = \frac{1}{s} \sum_{m=1}^s \frac{\tilde{X}_m}{X_{m0}}$$

$$\begin{aligned}
\text{Subject to } 1 &= \frac{1}{t} \sum_{n=1}^t \frac{\tilde{Y}_n}{Y_{n0}} \\
\tilde{X} &\geq \sum_{i \in I, i \neq 0}^n \pi_i X_i \\
\tilde{Y} &\leq \sum_{i \in I, i \neq 0}^n \pi_i Y_i \\
\tilde{X} &\geq z X_0 \quad \text{and} \quad \tilde{Y} \geq z Y_0 \\
\pi &\geq 0, \quad \tilde{Y} \geq 0, \quad z \geq 0
\end{aligned} \quad (5)$$

Nonradial slacks-based Malmquist productivity index. This article also employs the Malmquist TFP index to measure the impact of productivity change on the panel data. The concept of the Malmquist productivity index was first introduced by Malmquist (1953), and Caves *et al.* (1982) applied it in the nonparametric framework. Fare *et al.* (1994) developed it into the output-based Malmquist productivity change index. In their model, DEA can be applied to measure the productivity changes between two periods of a firm's activities in the specific set of DMUs. This approach is called the radial Malmquist productivity index. However, the radial model suffers from a neglect of slacks and creates an infeasible problem. In order to solve this problem, Zhu (1996), Chen (2003) and Cooper *et al.* (2004) developed a nonradial Malmquist productivity index. This article refers to Cooper *et al.* (2004) as the nonradial and slacks-based Malmquist productivity index to calculate a bank's productivity.

Following Cooper *et al.* (2004), the Malmquist index is computed as the product of (catch-up) and (frontier-shift). The catch-up term relates to the degree of DMU improving its efficiency, while the frontier-shift term reflects the Shift In Frontier Technology (SIFT) between the two time periods.

We now define the nonradial Malmquist productivity index by the nonradial efficiency scores of $\gamma_0^t(X_0^t, Y_0^t)$, $\gamma_0^{t+1}(X_0^{t+1}, Y_0^{t+1})$, $\gamma_0^{t+1}(X_0^t, Y_0^t)$ and $\gamma_0^t(X_0^{t+1}, Y_0^{t+1})$ as shown by:

$$\begin{aligned}
M^*(Y^1, X^1, Y^0, X^0) &= \frac{\gamma_0^t(X_0^t, Y_0^t)}{\gamma_0^{t+1}(X_0^{t+1}, Y_0^{t+1})} \\
&\times \left[\frac{\gamma_0^{t+1}(X_0^{t+1}, Y_0^{t+1})}{\gamma_0^t(X_0^t, Y_0^t)} \times \frac{\gamma_0^{t+1}(X_0^t, Y_0^t)}{\gamma_0^t(X_0^{t+1}, Y_0^{t+1})} \right]^{1/2}
\end{aligned} \quad (6)$$

The index M^* is in fact the geometric mean of two input-based Malmquist TFP indices. If M^* is greater than one, then this represents a positive TFP growth from period t_0 to period t_1 . The first term on the right-hand side of Equation 6 measures the technical efficiency change, which represents the Catching-up In Efficiency (CIE). The second term on the left-hand side of Equation 6 measures the technical change, which represents the SIFT between periods t_0 and t_1 . If CIE is larger than one, then the efficiency improvement has occurred within the two periods, but if CIE is smaller than one, then this indicates that bank efficiency has become worse than before. If SIFT is larger than one, then the production technology is progressive within two periods. However, the technology level is depressive if the value of SIFT is smaller than one.

IV. Empirical Results

The computer program DEA-Solver is employed to estimate the nonparametric programming model and Malmquist index.

The BCC and SBM super-efficiency indices – adjust the risk analysis

Concerning 'bad luck' hypotheses, 'bad management' hypotheses, 'skimping behaviour' hypotheses and 'moral hazard problems' hypotheses, we believe that banks with lower problem loans are tempted more to engage in higher efficiency operations. Higher problem loans result in banks purchasing additional input and paying more monitoring and controlling costs to solve the problem loans. Most scholars support this point of view. In other words, there is a negative correlation between problem loans and operating efficiency, implying that banks with lower problem loans also operate more efficiently. Hence, we expect that the efficiency score of high risk is lower than that for low risk in the BCC or SBM super-efficiency model.

The empirical results, based on DEA, are presented in Table 2 (Panels A and B). From the BCC estimation listed in Table 2 (Panel A), we find that the mean technical efficiency score of incorporating account risk (0.94587) is higher than that when not incorporating account risk (0.907825). After adjusting the risk factor, eight banks remain at the same rank, 16 banks, which have a lower degree of problem loans (i.e. the degree is less than 6.5 of the average degree), have increased their rank, while six banks with a lower degree of

Table 2. Results of technical efficiency assessment

Banks	TE (risk adjusted)	Ranking	TE (no risk)	Ranking	Problem loans
Panel A: by BCC (1998–2002)					
DMU1	0.90616	34	0.9054	20	7.048
DMU2	0.93734	27	0.93478	15	6.05
DMU3	0.93198	29	0.9247	18	6.048
DMU4	0.90334	36	0.84402	36	2.972
DMU5	0.92834	31	0.88914	26	8.392
DMU6	0.93434	28	0.88284	27	3.834
DMU7	0.9041	35	0.8428	37	9.278
DMU8	0.8527	41	0.8527	33	23.64
DMU9	1	1	1	1	18.232
DMU10	0.88372	38	0.84574	35	11.752
DMU11	0.968	17	0.92946	17	2.26
DMU12	0.89762	37	0.8827	28	8.96
DMU13	1	1	1	1	2.45
DMU14	0.99762	8	0.99692	6	3.366
DMU15	0.94372	25	0.88212	29	3.758
DMU16	0.9974	9	0.95544	11	2.644
DMU17	0.92958	30	0.90012	23	2.498
DMU18	0.97276	14	0.92426	19	4.756
DMU19	0.97256	15	0.97256	8	7.682
DMU20	1	1	0.96936	9	2.668
DMU21	0.91712	33	0.85034	34	4.692
DMU22	0.95242	23	0.83284	38	3.34
DMU23	0.95252	22	0.82448	40	1.366
DMU24	0.96956	16	0.89804	24	2.334
DMU25	0.9668	19	0.88098	30	2.776
DMU26	0.99714	10	0.9442	12	4
DMU27	0.94558	24	0.87978	31	2.624
DMU28	0.97492	13	0.9425	13	4.588
DMU29	0.87978	39	0.87498	32	13.762
DMU30	0.96226	20	0.90064	22	4.962
DMU31	0.99336	12	0.93578	14	3.912
DMU32	1	1	0.99338	7	12.482
DMU33	0.9409	26	0.78468	43	5.088
DMU34	0.85954	40	0.8328	39	4.096
DMU35	1	1	1	1	15.55
DMU36	0.81762	43	0.8028	42	14.796
DMU37	0.9954	11	0.93142	16	5.548
DMU38	0.96714	18	0.96714	10	9.096
DMU39	1	1	1	1	8.28
DMU40	1	1	1	1	4.05
DMU41	0.92064	32	0.9038	21	9.624
DMU42	0.96104	21	0.89788	25	5.994
DMU43	0.83766	42	0.82296	41	12.482
Total average	0.945876		0.907825		
Panel B: by SBM DEA-super (1998–2002)					
DMU1	0.566746	25	0.638756	22	7.048
DMU2	0.721345	16	0.776781	10	6.05
DMU3	0.633995	21	0.695349	19	6.048
DMU4	0.785103	11	0.773378	12	2.972
DMU5	0.490977	30	0.530856	29	8.392
DMU6	0.598115	23	0.603851	24	3.834
DMU7	0.211982	41	0.270139	41	9.278
DMU8	0.137847	43	0.182188	43	23.64
DMU9	0.204047	42	0.26716	42	18.232
DMU10	0.491652	29	0.565595	27	11.752
DMU11	1.112409	4	0.975511	5	2.26
DMU12	0.579014	24	0.657456	20	8.96
DMU13	1.489535	3	1.461306	2	2.45

(continued)

Table 2. Continued

Banks	TE (risk adjusted)	Ranking	TE (no risk)	Ranking	Problem loans
DMU14	1.065325	5	1.054465	3	3.366
DMU15	0.340677	36	0.361484	38	3.758
DMU16	102.4773	1	0.721916	17	2.644
DMU17	0.764364	12	0.737911	14	2.498
DMU18	0.786047	10	0.784998	7	4.756
DMU19	0.684312	19	0.733616	15	7.682
DMU20	0.994516	7	0.738282	13	2.668
DMU21	0.384879	35	0.438432	34	4.692
DMU22	0.448689	32	0.438012	35	3.34
DMU23	0.718078	17	0.573685	25	1.366
DMU24	0.660237	20	0.634983	23	2.334
DMU25	0.756577	13	0.780092	8	2.776
DMU26	0.827743	8	0.800371	6	4
DMU27	0.791113	9	0.775781	11	2.624
DMU28	0.752026	14	0.732668	16	4.588
DMU29	0.631797	22	0.702605	18	13.762
DMU30	0.562623	26	0.573316	26	4.962
DMU31	0.727906	15	0.648708	12	3.912
DMU32	0.717013	18	0.778567	9	12.482
DMU33	0.231901	40	0.288431	40	5.088
DMU34	0.455246	31	0.504724	13	4.096
DMU35	0.242494	39	0.319958	39	15.55
DMU36	0.285921	38	0.371032	37	14.796
DMU37	0.512691	27	0.543868	28	5.548
DMU38	0.417534	33	0.445358	33	9.096
DMU39	1.033054	6	1.036964	4	8.28
DMU40	1.76765	2	1.47709	1	4.05
DMU41	0.406619	34	0.456353	32	9.624
DMU42	0.49745	28	0.506058	30	5.994
DMU43	0.331945	37	0.423812	36	12.482
Total average	3.006894		0.64609		

problem loans have fallen down in their rank. For the banks with a higher degree of problem loans, 11 banks dropped in rank, while two banks increased their rank. In other words, 81% of banks changed their rank.

From the SBM super-efficiency estimation listed in Table 2 (Panel B), we find that the mean technical efficiency score of incorporating account risk (3.006894) is higher than that when not incorporating account risk (0.64609). After adjusting the risk factor, seven banks remain at the same rank, 14 banks with a lower degree of problem loans (i.e. the degree is less than 6.5 of the average degree) have increased their rank, while 11 banks with a lower degree of problem loans have fallen down in rank. For the banks with a higher degree of problem loans, 11 banks dropped in rank. In other words, 84% of banks changed their rank.

Based on the above results, it is noted that unless risk is controlled for, one might easily miscalculate a bank's level of inefficiency. In general, banks with a higher degree of problem loans dropped their efficiency by incorporating account risk, while banks

with a lower degree of problem loans increased their efficiency by incorporating risk. This shows that problem loans increase the expenses of dealing with these loans (such as extra input, monitoring, negotiating, etc.), which reduce a bank's efficiency. In other words, the efficiency score of high risk is lower than that for low risk. This result shows that risk factors have an impact on a bank's efficiency, which is consistent with Mester (1996), Altunbas *et al.* (2000), Hughes *et al.* (2001), Drake and Hall (2003), Girardone *et al.* (2004) and Ataullah *et al.* (2004).

In the SBM super-efficiency model, the incorporating accounting risks of DMUs 11, 13, 14, 16, 39 and 40 are efficient in the estimation. However, the super-efficiency method enables us to measure their efficiency with efficiency scores equal to one, leading to the following ranking units: DMUs 16 > 40 > 13 > 11 > 14 > 39. For not incorporating account risk, the DMUs of 13, 14, 39 and 40 are efficient in the estimation. However, the super-efficiency method leads to the following ranking units: DMUs 40 > 13 > 14 > 39.

Table 3. Results of the Malmquist index (1998–2002)

	Catch-up	Frontier-shift	Malmquist index
DMU1	0.883729	1.447412	1.279119
DMU2	1.313396	1.203399	1.58054
DMU3	0.638131	1.692244	1.079873
DMU4	0.765088	7.69837	5.889931
DMU5	0.939714	10.82417	10.17162
DMU6	0.199075	16.15507	3.216078
DMU7	0.025761	37.12391	0.956331
DMU8	4.77953	3.792163	18.12476
DMU9	0.888333	1.206676	1.071931
DMU10	0.534641	12.3612	6.608804
DMU11	2.105715	5.837058	12.29118
DMU12	0.664058	11.8667	7.880179
DMU13	1.124346	1.294163	1.455087
DMU14	0.72311	7.540307	5.452469
DMU15	0.41727	14.88804	6.212337
DMU16	0.050902	32.84014	1.671626
DMU17	0.596301	10.51797	6.271877
DMU18	0.306759	2.274602	0.697755
DMU19	1.318849	1.171814	1.545446
DMU20	0.998073	1.212836	1.210499
DMU21	0.639932	16.4145	10.50417
DMU22	0.069979	27.38078	1.916074
DMU23	0.428611	12.11457	5.192441
DMU24	0.285491	14.08673	4.02164
DMU25	1.077397	9.143254	9.850913
DMU26	0.510982	9.026949	4.612607
DMU27	0.426763	12.89414	5.502741
DMU28	1.240591	9.654577	11.97738
DMU29	1.86E-08	50	9.31E-07
DMU30	0.38961	13.6489	5.317742
DMU31	0.357224	16.77544	5.992593
DMU32	0.870807	1.362545	1.186513
DMU33	0.474567	19.58848	9.296045
DMU34	0.000424	36.54734	0.015481
DMU35	1.063973	1.180033	1.255524
DMU36	0.837363	11.92006	9.981415
DMU37	0.002641	37.26661	0.09841
DMU38	0.054887	1.372772	0.075348
DMU39	0.961066	0.240213	0.230861
DMU40	0.670416	0.339615	0.227683
DMU41	0.375497	18.51384	6.951897
DMU42	0.304431	14.52913	4.423122
DMU43	1.343663	11.95716	16.06639
Average	0.736258	12.30014	4.86894

Malmquist index analysis

Based on further calculations from Table 3, the average Malmquist index of all the banks is 4.86894, representing an increase in technical and technical efficiency of productivity. In analysing the components of the Malmquist index (4.86894), we find that the technical change index is 12.30014 and the technical efficiency change index is 0.736258, where the number 12.30014 of the technical change index implies that most of the growth in bank productivity in the prior year stemmed from technical progress

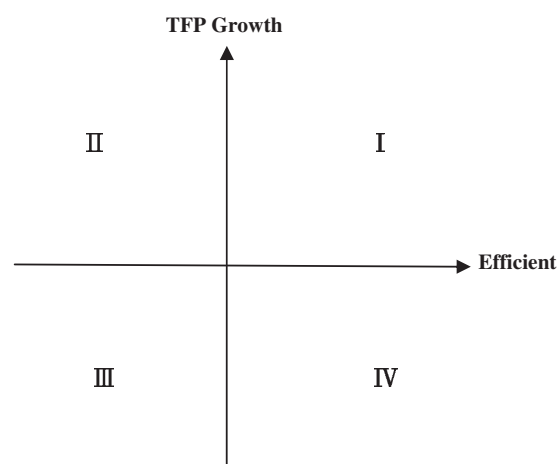


Fig. 1. The quadrant

Notes: (1) Quadrant I (efficient and positive TFP growth): Banks in this area possess efficiency and improved productivity. (2) Quadrant II (inefficient and positive TFP growth): Banks in this area possess inefficiency and positive TFP growth. (3) Quadrant III (inefficient and negative TFP growth): Banks in this area possess inefficiency and decreased productivity. (4) Quadrant IV (efficient and negative TFP growth): Banks in this area possess efficiency and decreased productivity.

and computerization. This is as a result of Taiwan's financial market being highly competitive and banks' production technology at an obvious growth mode. The technical efficiency index is 0.736358, showing that the technology in efficient banks has not improved.

Both the SBM super efficiency and Malmquist index analysis

From the above analysis, we can see that both efficiency and productivity are good indices in evaluating the performance of banks. Thus, we further use a quadrant to separate the banks into four groups with these two indices, as illustrated in Fig. 1. The horizontal axis is the efficient set, which means that banks located at above the horizontal axis are efficient. The vertical axis is the positive TFP growth, which means banks located to the right side of the vertical axis possess improved productivity.

From Table 4 we find that there are four banks in Quadrant I, and the related average problem loans of these banks are 2.68. There are 31 banks in Quadrant II, and the related average problem loans of these banks are 7.23. There are six banks in Quadrant III, and the related average problem loans of these banks are 7.756. Finally, there are two banks in Quadrant IV, and the related average efficiency scores problem loans of these banks are 6.165. From the above results, we find that the risks of these

Table 4. Problem loans and number of banks in the quadrant

	1998–2002		Risk
	No. of banks	Problem loans (avg.)	
Quadrant I	4	2.68	4
Quadrant II	31	7.23	2
Quadrant III	6	7.756	1
Quadrant IV	2	6.615	3

four types of banks are: Quadrant III > Quadrant II > Quadrant IV > Quadrant I. It shows that a bank with a higher performance will have less problem loans in Taiwan.

V. Conclusions

As the constraint of setting up a new bank has been eased by Taiwan's government in recent years, competition in the banking industry has risen due to a significant increase in the number of banks. The change in banks' customer focus from good performance enterprises to bad performance enterprises is considered to have high risk with a negative impact on bank performance. However, few studies treat risk as an important factor when analysing bank efficiency. This study uses the DEA approach, including BCC and SBM super-efficiency models, to investigate whether a bank's technical efficiency index results, incorporating account risk and not incorporating account risk, differ significantly. We also employ the nonradial and slacks-based Malmquist TFP index to measure the impact of productivity change on the panel data.

According to the data on 43 Taiwanese commercial banks for the 3-year period from 1998 to 2002 in the empirical results of Section V, we note the following: (1) risk factors have an impact on a bank's efficiency. In general, a bank with a higher degree of problem loans will see its efficiency drop down by incorporating account risk, but a bank with a lower degree of problem loans will see its efficiency increase after incorporating account risk. (2) super efficiency provides a good framework for ranking efficient units. (3) Based on the nonradial and slacks-based Malmquist TFP index, we find that bank productivity has increased and technical change has improved. (4) In Taiwan, banks with a higher performance will be more secure.

Based on the empirical results, it is very obvious as to the importance of incorporating risk into the

analysis of production when measuring a bank's technical efficiency in Taiwan's commercial banking industry. Consequently, a bank should carry out a careful investigation on business credit, financial condition, and previous bank history before lending to businesses since problem loans play such an important part of a bank's efficiency.

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Appendix

Summarization of the Bank Literature

Author(s)	Economies of scale and efficiency	Technique	Input/output variables	Risk (endogenous, exogenous, or excluded)
Hunter and Timme (1986)	Economies of scale and technical change	Stochastic frontier analysis	Input: (1) labour; (2) physical capital Output: (1) total loans; (2) total net loans and deposit; (3) total loans and securities less reserves for losses.	Risk: excluded
Berger <i>et al.</i> (1987)	Economies of scale and scope	Iterative seemingly unrelated cost regression model	Input: (1) labour; (2) office space Output: (1) real estate loans; (2) commercial loans; (3) installment loans; (4) time and savings deposits; (5) demand deposits.	Risk: excluded
Ferrier and Lovell (1990)	Efficiency	(1) Stochastic frontier analysis (2) BCC	Input: (1) labour; (2) occupancy costs and expenditure on furniture and equipment; (3) expenditure on materials Output: (1) real estate loans; (2) commercial loans; (3) installment loans; (4) demand deposits; (5) time demand deposits.	Risk: excluded
Aly <i>et al.</i> (1990)	Allocative, scale and technical efficiency	BCC	Input: (1) labour; (2) capital; (3) loanable funds Output: (1) real estate loans; (2) commercial and industrial loans; (3) consumer loans; (4) all other loans; (5) demand deposits.	Risk: excluded
Berger and Humphrey (1991)	(1) Economies of scale and scope (2) Allocative, scale and technical efficiency	Stochastic frontier analysis	Input: (1) labour; (2) capital Output: (1) real estate loans; (2) commercial and industrial loans; (3) installment loans; (4) time and savings deposits; (5) demand deposits.	Risk: excluded
Cebenoyan <i>et al.</i> (1993)	Efficiency	(2) Two stage	Input: (1) labour; (2) physical capital; (3) deposit Output: (1) mortgage lending; (2) nonmortgage loans; (3) other investment.	Risk: excluded
Grabowski <i>et al.</i> (1993)	Allocative, scale and technical efficiency	Stochastic frontier analysis	Input: (1) labour; (2) capital; (3) loan funds Output: (1) real estate loans; (2) commercial and industrial loans; (3) consume loans; (4) demand deposit; (5) investment securities.	Risk: excluded
McAllister and McManus (1993)	Allocative, scale and technical efficiency	Stochastic frontier analysis	Input: (1) labour; (2) capital; (3) purchased funds Output: (1) real estate loans; (2) commercial and industrial loans; (3) installment loans; (4) saving deposits; (5) demand deposits.	Risk: excluded

(continued)

Continued

Author(s)	Economies of scale and efficiency	Technique	Input/output variables	Risk (endogenous, exogenous, or excluded)
Barr <i>et al.</i> (1994)	Efficiency	(1) CCR (2) Two stage	Input: (1) employee salary; (2) assets; (3) other noninterest expenses; (4) total interest expense; (5) purchased funds Output: (1) core deposits; (2) earning assets; (3) total interest income.	Risk: exogenous
Elyasiani <i>et al.</i> (1994)	Allocative and technical efficiency	(1) BCC (2) Two stage	Input: (1) labour; (2) capital; (3) loan funds Output: (1) real estate loans; (2) commercial and industrial loans; (3) consume loans; (4) demand deposit; (5) securities.	Risk: excluded
Favero and Papi (1995)	Allocative, scale and technical efficiency	(1) BCC (2) Two stage	Input: (1) labour; (2) capital; (3) loanable funds; (4) net funds Output: (1) loans; (2) investment; (3) noninterest income.	Risk: excluded
Zaim (1995)	Allocative and technical efficiency	(1) CCR (2) BCC	Input: (1) labour; (2) interest expenditures; (3) depreciation expenditures; (4) materials expenditures Output: (1) long-term loans; (2) short loans; (3) time deposits; (4) demand deposits.	Risk: excluded
Mester (1996)	Economies of scale and scope	Stochastic cost frontier	Input: (1) labour; (2) physical capital; (3) deposit to produce earning asset Output: (1) real estate loans; (2) commercial and industrial loan; (3) loans to individuals.	Risk: endogenous
Berger and DeYoung (1997)	X-efficiency	Fourierflexible stochastic cost frontier	Input: (1) labour; (2) physical capital Output: (1) consumer loans; (2) commercial loans; (3) real estate loans; (4) transaction deposits; (5) fee-based income.	Risk: exogenous
Bhattacharyya <i>et al.</i> (1997)	Efficiency	(1) Stochastic frontier analysis (2) BCC	Input: (1) labour; (2) capital; (3) other nonfinancial inputs Output: (1) deposits; (2) physical quantities of advances.	Risk: excluded
Resti (1997)	Allocative efficiency and technical efficiency	(1) Stochastic frontier analysis (2) BCC	Input: (1) labour; (2) capital Output: (1) loans; (2) deposits; (3) noninterest income	Risk: excluded

Schaffnit <i>et al.</i> (1997)	Allocative efficiency and technical efficiency	DEA AR	Input: (1) tellers; (2) typing; (3) accounting; (4) supervision; (5) credit Output: (1) personal loans; (2) commercial loans; (3) deposit sales; (4) counter transactions; (5) security transactions; (6) counter sales.	Risk: excluded
Deyoung and Hasan (1998)	Allocative efficiency and technical efficiency	(1) Fourierflexible stochastic profit frontier (2) Two stage	Input: (1) labour; (2) physical capital; (3) funds Output: (1) total loans; (2) transaction deposits; (3) fee-based financial service.	Risk: excluded
Chang (1999)	Technical efficiency	BCC	Input: (1) labour; (2) capital; (3) loanable funds Output: (1) ordinary loans; (2) government subsidized loans; (3) deposits; (4) other noninterest income.	Risk: endogenous
Fukuyama <i>et al.</i> (1999)	(1) Scale and technical efficiency (2) Productivity	BCC	Input: (1) labour; (2) assets; (3) deposits Output: (1) loans; (2) investment.	Risk: excluded
Hughes (1999)	Economies of scale	Stochastic cost	Input: (1) labour; (2) physical capital; (3) insured deposits; (4) other borrowed money; (5) insured deposits Output: (1) loans to individuals; (2) commercial and industries loans; (3) real estate loans; (4) other loans; (5) securities.	Risk: endogenous
Pastor (1999)	Risk management efficiency	DEA incorporated into environmental variables	Input: (1) financial cost; (2) operating expenses Output: (1) financial revenues; (2) stock and bond portfolio returns.	Risk: endogenous
Altunbus <i>et al.</i> (2000)	Economies of scale and efficiency	Fourierflexible stochastic cost frontier	Input: (1) labour; (2) physical capital; (3) funds Output: (1) total loans; (2) total securities; (3) off balance sheet.	Risk: exogenous
Chen and Yeh (2000)	(1) Scale and technical efficiency (2) Productivity	(1) Stochastic frontier analysis (2) BCC	Input: (1) labour; (2) assets; (3) deposits Output: (1) loans; (2) investment; (3) noninterest income.	Risk: excluded
Huang and Wang (2002)	Technical efficiency	(1) Stochastic cost frontier (2) Fixed fee approach (3) BCC (4) Fourierflexible stochastic cost frontier	Input: (1) Labour; (2) physical capital; (3) deposit Output: (1) short-term loans; (2) long-term loans; (3) investment.	Risk: excluded
Hughes <i>et al.</i> (2000)	Economies of scale	Stochastic cost and profit frontier	Input: (1) labour; (2) physical capital; (3) insured deposits; (4) other borrowed money; (5) insured deposits Output: (1) loans to individuals; (2) commercial and industries loans; (3) real estate loans; (4) other loans; (5) securities.	Risk: endogenous

(continued)

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Author(s)	Economies of scale and efficiency	Technique	Input/output variables	Risk (endogenous, exogenous, or excluded)
Hughes <i>et al.</i> (2001)	Economies of scale	Stochastic cost and profit frontier	Input: (1) labour; (2) physical capital Output: (1) liquid assets; (2) short-term securities; (3) short-term securities; (4) loans and leases net of unearned income; (5) other assets.	Risk: endogenous
Casu and Molyneux (2003)	Efficiency analysis	(1) Two stage: BCC and Tobit regression	Input: (1) total cost; (2) total deposits Output: (1) total loans; (2) other earned assets.	Risk: excluded
Drake and Hall (2003)	Scale and technical efficiency	BCC	Input: (1) general and administrative expenses; (2) fixed asset; (3) retail and wholesale deposits Output: (1) loans; (2) liquid assets; (3) other income	Risk: exogenous
Girardone <i>et al.</i> (2004)	Economies of scale and X-efficiency	Fourierflexible stochastic cost frontier	Input: (1) labour; (2) fixed assets; (3) deposits Output: (1) total customer loans; (2) other earning assets.	Risk: endogenous
Ataullah <i>et al.</i> (2004)	Scale and pure technical efficiency	BCC	Input: (1) labour; (2) physical capital Output: (1) consumer loans; (2) commercial loans; (3) real estate loans; (4) transaction deposits; (5) fee-based income.	Risk: exogenous